GURU NANAK DEV ENGINEERING COLLEGE, BIDAR
Department of Computer Science & Engineering | VTU, Belagavi

A Web Portal for Academic Project Lifecycle Management

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Introduction

Background

Academic projects are a vital part of engineering education. However, most institutions still rely on manual methods — Google Forms, spreadsheets — leading to:

- Duplicate project titles across batches
- Inefficient and manual guide allocation
- Scattered records with no easy retrieval

Proposed Solution

This project extends the existing evaluation system built by our seniors into a full-fledged Web Portal for Academic Project Lifecycle Management — covering everything from title submission and guide allocation to synopsis approval, progress tracking, and evaluation.

Phase 1

Project Proposal & Title Submission

Phase 2

Guide Allocation & Progress Tracking

Phase 3

Synopsis upload & approval

Phase 4

Evaluation & Report Generation

Problem Statement

Current Challenges in Academic Project Management

Duplicate Project Titles

No mechanism to detect if a project title has already been used in previous batches, leading to repeated work.

Manual Guide Allocation

Manually pairing a large volume of student groups with faculty based on specific technical domains is time-consuming and prone to misalignment.

No Centralized Tracking

Project status is tracked using spreadsheets, making it impossible to get a real-time overview of all projects.

Scattered Records & Data Loss

Historical project records are difficult to maintain and often lost, making future reference nearly impossible.

Student Confusion

Students face confusion about submission deadlines, approval status, and evaluation results with no single source of truth.

Faculty Overhead

Teachers must manually manage multiple groups, maintain separate records.

Motivation

What inspired us to work on this project?

01

Solve a Real Problem

We didn't need to look far. Project lifecycle management usually works fully manual — Google Forms for submissions, WhatsApp for approvals, and Excel sheets for records. We saw an opportunity to build something that would directly benefit our own institution and future batches.

02

Saw Our Seniors Struggle with Manual Project Tracking

We watched our seniors deal with duplicate project titles going undetected and students having no clear way to track their synopsis approval or evaluation status. These were real, recurring problems that deserved a proper solution.

03

Extend & Build On the Existing PMES System

Our seniors built Project Management and Evaluation System (PMES) — a solid foundation for evaluation management. But the project lifecycle starts long before evaluation: title submission, duplicate checks, synopsis approval, guide allocation. We were motivated to complete the picture by extending PMES into a full lifecycle portal.

Literature Review

S. No	Title	Author	Year	Techniques	Key Findings
1	A Web-Based Academic Project Mgmt & Evaluation System	S. S. Pandey et al.	2025	Role-based Flask app; digital submission	Enabled digital project submission; reduced paperwork
2	Student Performance Evaluation System	A. D. Shirale & K. T. Mali	2024	Data analytics + ML	Automated tracking; improved evaluation accuracy
3	Web-Based Projects Evaluation Mgmt System	M. O. Oyelami et al.	2008	Online evaluation forms	Digitized evaluation; saved time and improved access
4	Student Project-Mgmt & Evaluation Strategies	W. K. Wanyoike et al.	2023	Survey + requirement analysis	Gaps in tools; recommended integrated platforms
5	Quality Evaluation System for Higher Education	C. L. T. Dang & S. Zhang	2024	Statistical modeling + survey	Multi-factor evaluation; high feasibility validated

Literature Review (Continued)

S. No	Title	Author	Year	Techniques	Key Findings
6	Distributed Student Performance Assessment	A. Montoya & D. Jiménez	2024	Agile dev + DB replication	Transparent assessment with real-time monitoring
7	Academic Project Nexus: Final-Year Mgmt System	R. A. Helonde et al.	2024	Web app (HTML, CSS, MySQL)	Centralized repo; secure documentation & collaboration
8	Responsive Web-Based Final-Year Project Repo	O. I. Uzugogbo & A. M. Kolade	2024	Laravel + usability testing	Simplified submission; high user adoption rate
9	University Results Extraction & Analysis Tool	G. Geethalaxmi et al.	2022	Node.js + Python + OpenPyXL	Automated Excel reports; reduced manual entry errors
10	Electronic Learner Data Management System	M. N. De Guzman et al.	2024	SDLC + Agile; TAM evaluation	Improved data accuracy; strong user acceptance

Inference & Gap Identification

Inferences from Literature Review

1. Most systems focus on limited features (only marks storage or only submission) — not full lifecycle.
2. Automated guide allocation based on domain expertise is largely unexplored.
3. Duplicate project title detection is absent in nearly all existing tools.
4. Centralized tracking from proposal to evaluation is missing.
5. Role-based access control improves transparency and reduces admin load.

Research Gaps Addressed by This Project

No Full Lifecycle Coverage

This portal covers title submission to evaluation.

No Duplicate Detection

Adding a past-project repository to prevent repeated project ideas.

No Smart Guide Assignment

Guide allocation is automated based on domain expertise.

No Integrated Synopsis Workflow

Synopsis upload, review, and approval with feedback.

Poor Reporting

Auto-generated PDF reports for every review phase are included.

Project Objectives

Main Objective

To develop a centralized web portal that manages the entire lifecycle of academic projects — from proposal to evaluation — building upon and extending the existing PMES developed by the previous batch.

01

Centralized Portal

Develop a unified web platform supporting students, teachers, and admin roles.

02

Title Submission

Allow students to select project domains and submit project titles online.

03

Guide Allocation

Automate guide assignment based on domain expertise and workload.

04

Synopsis Approval

Implement a structured synopsis upload, review, and approval mechanism.

05

Duplicate Detection

Maintain a past-project repository to prevent duplication of project titles.

06

Evaluation System

Enable teachers to evaluate projects across 4 review phases (Phase 1 & 2).

Proposed Methodology

3-Layer Client-Server Architecture

Presentation Layer	**HTML • CSS • JavaScript • Jinja2 Templates** Student, Teacher & Admin dashboards — project submission, status tracking, review forms
Application Layer	**Python • Flask Framework • SQLAlchemy ORM** Authentication, title submission, synopsis management, guide allocation, evaluation processing, report generation
Data Layer	**SQLite (dev) • MySQL (production) • File Storage** User records, project proposals, synopsis docs, evaluation marks, past project repository

Software & Technology Stack

Frontend

HTML5

CSS3

JavaScript

Jinja2 Templates

Backend

Python 3.x

Flask 3.0

SQLAlchemy 2.0

Flask-Login

Database

SQLite (Development)

MySQL (Production)

PyMySQL

Libraries & Tools

OpenPyXL – Excel processing

ReportLab – PDF generation

python-dotenv – Config mgmt

openpyxl – Excel import

Expected Outcome

The proposed portal will deliver:

01

Fully Functional Web Portal

A complete, role-based web portal managing the entire academic project lifecycle end-to-end.

02

Automated Synopsis Workflow

Students submit synopses, teachers review and approve/reject with structured feedback online.

03

Duplicate Title Prevention

A searchable repository of past project titles prevents repetition across batches.

04

Smart Guide Allocation

Guide assignment based on domain expertise ensures better mentor-student compatibility.

05

4-Phase Evaluation System

Extends the senior PMES to support Phase 1 & 2 reviews with per-evaluator marks and averaging.

06

Report Generation

Auto-generated PDFs and Excel exports for individual students and batch-level summaries.

References

[1] Pressman, R. S. (2014). Software Engineering: A Practitioner's Approach. McGraw-Hill.

[2] Sommerville, I. (2016). Software Engineering (10th ed.). Pearson Education.

[3] Flask Documentation – https://flask.palletsprojects.com

[4] SQLAlchemy Documentation – https://www.sqlalchemy.org

[5] S. S. Pandey et al., "A Web-Based Academic Project Management and Evaluation System," IRJMETS, 2025.

[6] M. O. Oyelami & J. B. Odajuyigbe, "Web-Based Projects Evaluation Management System," IEEE, 2008.

[7] R. A. Helonde et al., "Academic Project Nexus: Final-Year Project Management System," IRJMETS, 2024.

[8] O. I. Uzugogbo & A. M. Kolade, "Responsive Web-Based Final-Year Project Repository," WJAETS, 2024.

[9] G. Geethalaxmi et al., "University Results Extraction & Analysis Tool," IJAEM, 2022.

[10] M. N. De Guzman et al., "Electronic Learner Data Management System," IJRISS, 2024.

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Thank You!

We welcome your questions and feedback.

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